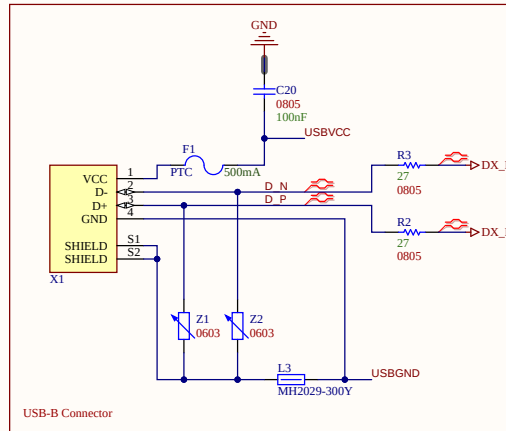
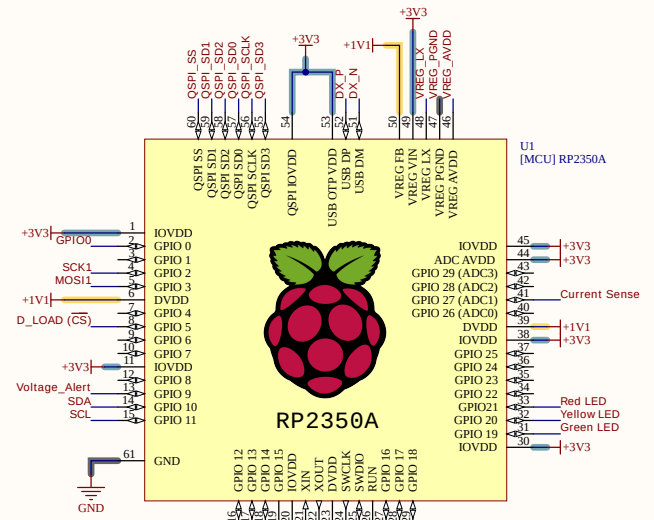
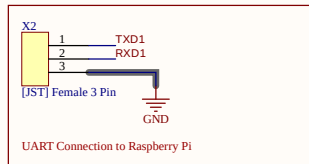
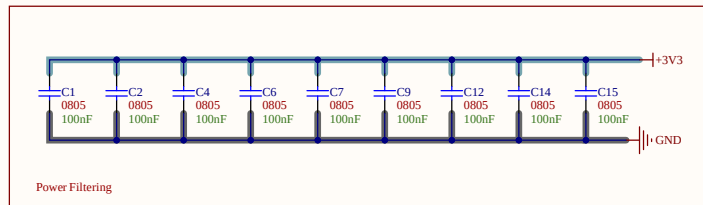
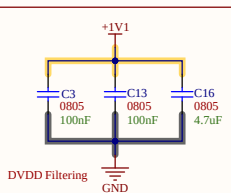
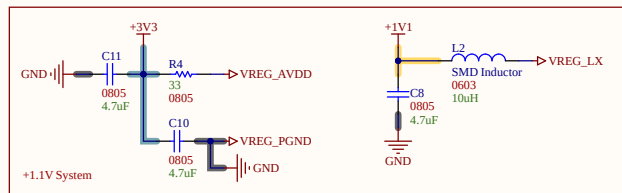
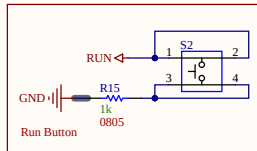
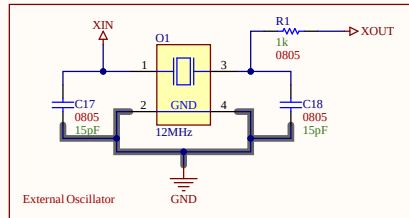
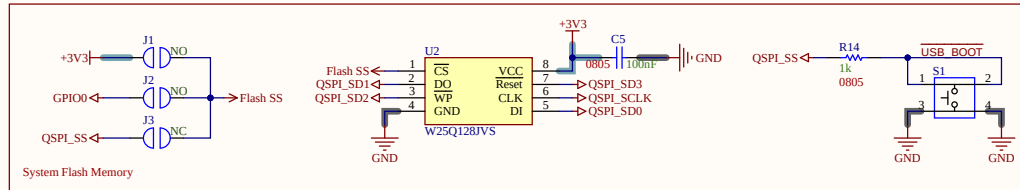
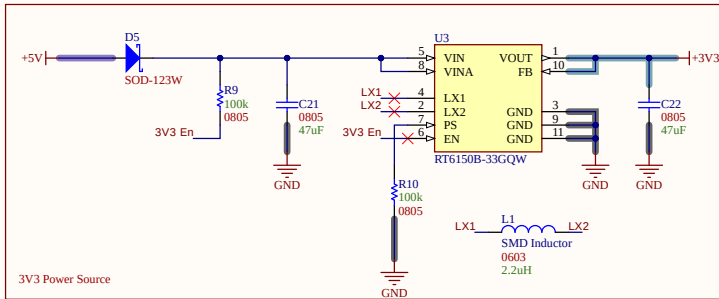
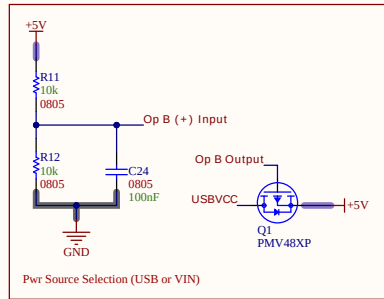
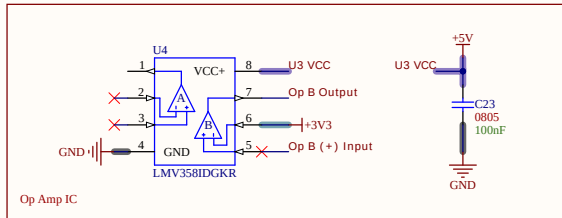
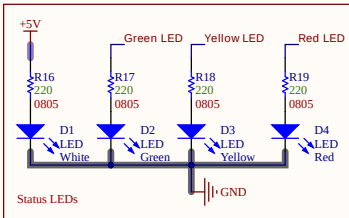
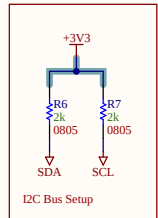
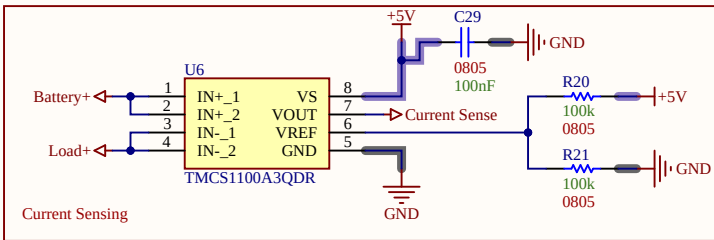
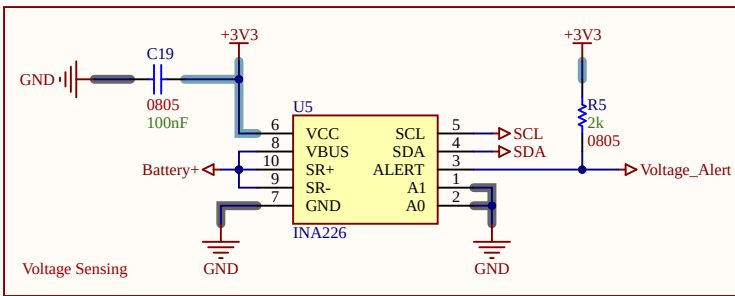
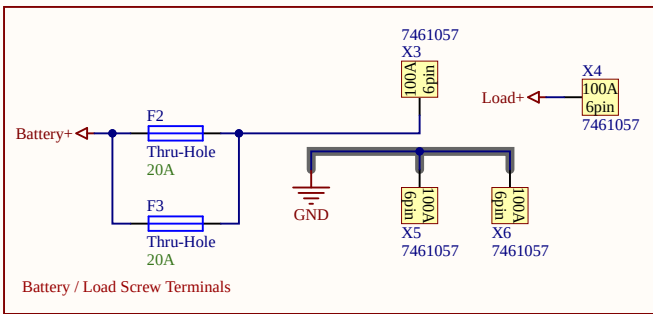




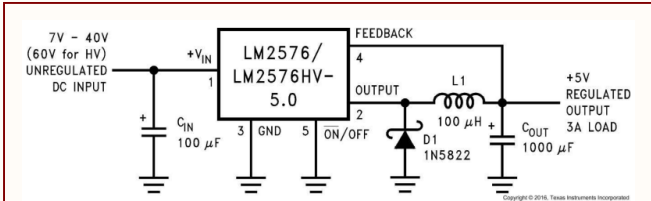
Project/Vehicle: Battery Monitor V1.PrjPcb		Machine Intelligence Laboratory	
Author(s): -Russell MacGregor		This is the main MCU for the Battery Monitor Board.	
-		Git Repo: BatteryMonitorSpring2025	
-		Git Hash: 3d1eaa94	
Date: 09/10/25	Revision: 0	Size: A3	File: Primary MCU.SchDoc



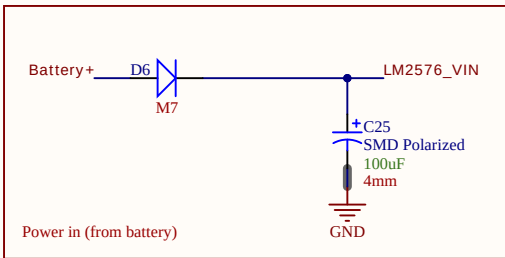


Project/Vehicle: Battery Monitor V1.PrjPcb			Machine Intelligence Laboratory This IC monitors the voltage of the battery and the current drawn from it.			
Author(s): -Russell MacGregor		Revisor(s): -				
-		-				
-		-				
Date: 09/10/25		Revision:		Size: A4		
File: Battery Monitoring.SchDoc		Sheet 2 of 4				

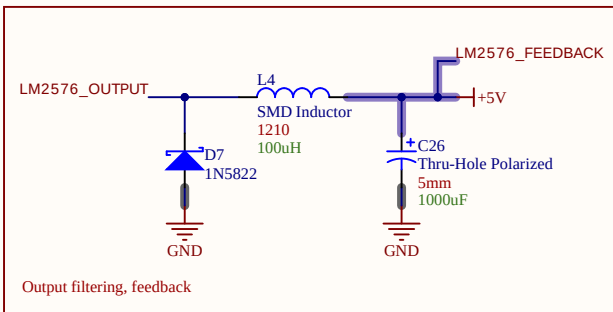
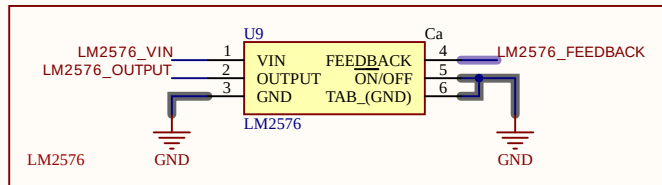




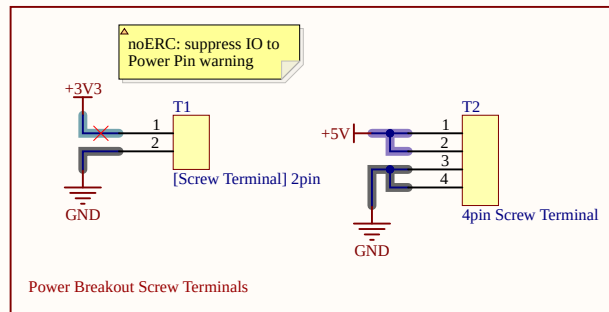
Circuit to implement



Power in (from battery)



Output filtering, feedback



Power Breakout Screw Terminals

Project/Vehicle: Battery Monitor V1.PrjPcb			Machine Intelligence Laboratory This steps down the ~22V from the battery to 5V for use with PCB electronics.		
Author(s): -Russell MacGregor - -		Revisor(s): - - -			
Date: 09/10/25			Git Repo: BatteryMonitorSpring2025 Git Hash: 3d1eaa94		
Date: 09/10/25	Revision: *	Size: A4	File: DC Step-Down Converter.SchDoc	Sheet 4 of 4	

Voltage

Current

SEG1

D1 D2 D3 D4

4-Digit 7seg Display
(Common Cathode)

E DP C G D4

UART

X2

D4 D3 D2 D1
Error Warning Okay On

D7

Battery Monitor
RADM
U1

Max Current
80A

C26

Load-

Load+

Batt+

Batt-

X4

F2

F3

X3

X5

X6

T1 +3V3
[Front]
2-Pin Screw Term

X1 Port

T2 +5V
[Front]
4-Pin Screw Term

